

Cayley, Collected Mathematical Papers, Vol. X

Pages 526–550 (scan); paper 704: A Memoir on the Single and Double Theta-Functions.

[Continuation of Paper 704: A Memoir on the Single and Double Theta-Functions.]

[Scan p. 506]

that is,

c^2	c^2	c^2	c^2	c^2	c^2
9	2	+ 4	15	− 3	8
2	12	− 15	1	− 8	6
12	9	− 1	4		
				+ 6	3
		− 4	3		
1	6			+ 15	8
6	12	+ 3	9	− 8	2
12	1				
		− 9	4	− 2	15
− 0	12	+ 4	8	+ 3	15
− 0	1	+ 3	2	+ 8	9
− 0	6	− 9	15	+ 2	4
− 0	9	− 15	6	+ 8	1
+ 0	4	− 8	12	− 6	2
− 0	3	− 12	15	− 1	2
− 0	2	+ 1	3	+ 4	6
+ 0	10	+ 6	9	− 3	12
+ 0	8	− 12	4	− 9	1

= 0;

each of the first set of 15 giving a homogeneous linear relation between four terms c^4 ; and each of the second set giving a homogeneous linear relation between three terms $c^2.c^2$, formed with the 10 constants c . Thus the first equation is

$$c_{92}^4 + c_{12}^4 + c_4^4 - c_8^4 = 0;$$

and so for the other lines of the two diagrams.

79. I form in the two notations the following tables:

TABLE OF THE 16 KUMMER HEXADS.

<i>A</i>	<i>A</i>	<i>A</i>	<i>A</i>	<i>B</i>	<i>B</i>	<i>B</i>	<i>B</i>	<i>C</i>	<i>C</i>	<i>C</i>	<i>D</i>	<i>D</i>	<i>E</i>	<i>A</i>
<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>E</i>	<i>F</i>	<i>F</i>	<i>B</i>
<i>AB</i>	<i>AC</i>	<i>AD</i>	<i>AE</i>	<i>BC</i>	<i>BD</i>	<i>BE</i>	<i>AB</i>	<i>CD</i>	<i>CE</i>	<i>AC</i>	<i>DE</i>	<i>AD</i>	<i>AE</i>	<i>C</i>
<i>CD</i>	<i>BD</i>	<i>BC</i>	<i>BC</i>	<i>AC</i>	<i>AD</i>	<i>AC</i>	<i>AC</i>	<i>AB</i>	<i>AB</i>	<i>BC</i>	<i>AB</i>	<i>BD</i>	<i>BE</i>	<i>D</i>
<i>CE</i>	<i>BE</i>	<i>BE</i>	<i>BD</i>	<i>AD</i>	<i>AE</i>	<i>AE</i>	<i>AD</i>	<i>BD</i>	<i>AE</i>	<i>AD</i>	<i>CD</i>	<i>AC</i>	<i>CD</i>	<i>E</i>
<i>DE</i>	<i>DE</i>	<i>CE</i>	<i>CD</i>	<i>AE</i>	<i>DE</i>	<i>CE</i>	<i>CD</i>	<i>BE</i>	<i>BE</i>	<i>BD</i>	<i>CE</i>	<i>BC</i>	<i>DE</i>	<i>F</i>

11	11	11	11	7	7	7	7	5	5	5	13	13	14	11
7	5	13	14	10	5	13	14	10	13	14	10	14	10	7
6	4	14	12	6	8	3	6	2	1	4	9	12	15	5
2	8	8	4	12	4	4	8	6	6	8	6	3	13	2
1	3	3	3	12	15	15	12	15	12	2	4	2	1	14
9	9	1	2	15	9	1	2	3	3	1	8	9	9	10

[Scan p. 507] *Table of the 120 pairs.* A large two-page tabular display in two parallel notations (a letter-pair notation and a numerical notation), the page being printed sideways. Each row gives an unordered pair of indices, the $15 \times 16/2 = 120$ pairs being arranged in 8 columns of 15 rows in each notation.

[Figure: full sideways table, letter-pair notation (8 columns), reproduced from scan as printed. The pairs run, in the letter notation, *AB.BE, AC.CE, AD.DE, AE.BD, AC.CE, AD.DE, BC.CE, BD.BE, ...*; in the numerical notation, the corresponding entries are 11.7, 11.5, 11.13, 11.14, 11.10, 6.4, 6.3, See scan for the full layout.]

[Scan p. 508]

81. Table of the 60 Göpel tetrads.

<i>A.B.AE.BE</i>	<i>C.D.CE.DE</i>	<i>E.F.AB.CD</i>	<i>AC.BD.AD.BC</i>
<i>A.B.AD.BD</i>		<i>D.F.AB.CE</i>	<i>AC.BE.AE.BC</i>
<i>A.B.AC.BC</i>	<i>CE.CD.DE</i>	<i>D.E.CD.CE</i>	<i>AD.BE.AE.BD</i>
<i>A.C.AE.CE</i>	<i>CF.AB.DE</i>	<i>E.F.AC.BD</i>	<i>AB.CD.AD.BC</i>
<i>A.C.AD.CD</i>		<i>D.F.AC.BE</i>	<i>AB.CE.AE.BC</i>
<i>A.C.AB.BC</i>	<i>B.D.BE.DE</i>	<i>D.E.BD.BE</i>	<i>AD.CE.AE.CD</i>
	<i>B.E.BD.DE</i>		
<i>A.D.AE.DE</i>	<i>B.F.AC.DE</i>	<i>E.F.AD.BC</i>	<i>AB.CD.AC.BD</i>
<i>A.D.AC.CD</i>			<i>AB.DE.AE.BD</i>
<i>A.D.AB.BD</i>	<i>B.C.BE.CE</i>	<i>C.F.AD.BE</i>	<i>AC.DE.AE.CD</i>
	<i>B.E.BC.CE</i>	<i>C.E.CD.DE</i>	
<i>A.E.AD.DE</i>	<i>B.F.AD.CE</i>		<i>AB.CE.AC.BE</i>
<i>A.E.AC.CE</i>		<i>D.F.AE.BC</i>	<i>AB.DE.AD.BE</i>
<i>A.E.AB.BE</i>	<i>B.C.BD.CD</i>	<i>C.F.AE.BD</i>	<i>AC.DE.AD.CE</i>
	<i>B.D.BC.CD</i>	<i>C.D.BC.BD</i>	
<i>A.F.BC.DE</i>	<i>B.F.AE.CD</i>		<i>BD.CE.BE.CD</i>
<i>A.F.BD.CE</i>		<i>D.E.AD.AE</i>	<i>BC.DE.BE.CD</i>
<i>A.F.BE.CD</i>	<i>B.C.AB.AC</i>		<i>BC.DE.BD.CE</i>
		<i>C.E.AC.AE</i>	
	<i>B.D.AB.AD</i>	<i>C.D.AC.AD</i>	
	<i>B.E.AB.AE</i>		

The same tetrads in the numerical notation:

11 7 15 3	5 13 1 9	14 10 6 2	4 12 8
11 7 12 4	5 14 2 9	13 10 6 1	
11 7 4 8	5 10 6 9	13 14 2 1	4 3 15 8
			12 3 15
11 5 15 1	7 13 3 9	14 10 4 3	6 2 12 8
11 5 12 2			6 1 15 8
11 5 6 8	7 14 4 9	13 10 4 3	12 1 15 2
	7 10 4 9	13 14 3 3	
11 13 15 9	7 5 3 1	14 10 12 8	6 2 4
11 13 4 2	7 14 8 1	5 10 12 3	
11 13 6 3	7 10 12 1	5 14 2 9	6 9 15
			4 9 15 2
11 14 12 9	7 5 4 2	13 10 15 8	6 1 4 3
11 14 4 1		5 10 15	
11 14 6 3	7 13 8 2	5 13 8	6 9 12 3
	7 10 16 2		4 9 12 1
11 10 8 9	7 5 6 4	13 14 12 15	1 3 2
11 10	7 13 6 12	5 14 4 15	3 2
11 10 1	7 14 6 15	5 13 4 12	8 9
			8 9 1
			3 2

[Scan p. 509]

The product-theorem, and its results.

82. The product-theorem was

$$\Theta\left(\begin{smallmatrix}\alpha,\beta\\ \gamma,\delta\end{smallmatrix}\right)(u+u')\cdot\Theta\left(\begin{smallmatrix}\alpha',\beta'\\ \gamma',\delta'\end{smallmatrix}\right)(u-u')=\Sigma\Theta\left(\begin{smallmatrix}\frac{1}{2}(\alpha+\alpha')+p,\frac{1}{2}(\beta+\beta')+q\\ \gamma+\gamma',\delta+\delta'\end{smallmatrix}\right)(2u)\cdot\Theta\left(\begin{smallmatrix}\frac{1}{2}(\alpha-\alpha')+p,\frac{1}{2}(\beta-\beta')+q\\ \gamma-\gamma',\delta-\delta'\end{smallmatrix}\right)(2u'),$$

where only one argument is exhibited, viz. $u+u'$, $u-u'$, $2u$, $2u'$ are written in place of $(u+u', v+v')$, $(u-u', v-v')$, $(2u, 2v)$, $(2u', 2v')$ respectively. The expression on the right-hand side is always a sum of

four terms, corresponding to the values $(0, 0)$, $(1, 0)$, $(0, 1)$, and $(1, 1)$ of (p, q) . For the development of the results it was found convenient to use the following auxiliary diagram.

Upper half of characteristic.

[Figure: large 16×16 table indexed by pairs of upper-half-characteristic combinations, exhibiting for each pair $(\frac{\alpha+\alpha'}{2}+p, \frac{\beta+\beta'}{2}+q)$ versus $(\frac{\alpha-\alpha'}{2}+p, \frac{\beta-\beta'}{2}+q)$ the upper indices, written as fractions, $0, 1, \frac{1}{2}, \frac{3}{2}$, with the $\pm$ signs of the four terms. Reproduced as printed in the scan; the cells contain entries of the form $\frac{1}{2}1, 11, \frac{3}{2}1$ etc., with leading signs to be combined with the lower-half diagram on the facing page.]

[Scan p. 510]

Lower half of characteristic.

[Figure: companion 16×16 table for the lower-half characteristics, structured exactly as the previous one; the entries are the values $\frac{\gamma+\gamma'}{2}, \frac{\delta+\delta'}{2}$ versus $\frac{\gamma-\gamma'}{2}, \frac{\delta-\delta'}{2}$, taking the values $0, 1, \frac{1}{2}, -\frac{1}{2}$, with the corresponding $\pm$ signs to be combined with the upper-half diagram.]

83. The upper characters of the Θ 's have thus the values $0, 1, \frac{1}{2}, \frac{3}{2}$; the lower characters are originally $2, 1, 0$, or -1 , and these have when necessary to be, by the addition or subtraction of 2 , reduced to 0 or 1 ; the effect of this change is either to leave the Θ unaltered, or to multiply it by -1 or $+i$, as follows:

$$\begin{aligned} \Theta\left(\begin{smallmatrix} 0 \\ \gamma \end{smallmatrix}\right) - 2 &= \Theta\left(\begin{smallmatrix} 0 \\ \gamma \end{smallmatrix}\right), & \Theta\left(\begin{smallmatrix} \frac{1}{2} \\ \gamma \end{smallmatrix}\right) + 2 &= i \Theta\left(\begin{smallmatrix} \frac{1}{2} \\ \gamma \end{smallmatrix}\right), & \Theta\left(\begin{smallmatrix} \frac{3}{2} \\ \gamma \end{smallmatrix}\right) - 2 &= -i \Theta\left(\begin{smallmatrix} \frac{3}{2} \\ \gamma \end{smallmatrix}\right), \\ \Theta\left(\begin{smallmatrix} 1 \\ \gamma \end{smallmatrix}\right) - 2 &= -\Theta\left(\begin{smallmatrix} 1 \\ \gamma \end{smallmatrix}\right), & \Theta\left(\begin{smallmatrix} \frac{1}{2} \\ \gamma \end{smallmatrix}\right) - 2 &= -i \Theta\left(\begin{smallmatrix} \frac{1}{2} \\ \gamma \end{smallmatrix}\right), & \Theta\left(\begin{smallmatrix} \frac{3}{2} \\ \gamma \end{smallmatrix}\right) + 2 &= i \Theta\left(\begin{smallmatrix} \frac{3}{2} \\ \gamma \end{smallmatrix}\right), \end{aligned}$$

where only the first column of characters is shown, but the same rule applies to the second column; and where we must of course combine the multipliers corresponding to the first and second columns respectively: for instance

$$\Theta\left(\begin{smallmatrix} \frac{3}{2} \\ \gamma \end{smallmatrix}\right) + 2 \cdot \Theta\left(\begin{smallmatrix} \frac{1}{2} \\ \gamma \end{smallmatrix}\right) + 2 = (-i \cdot -i) \Theta\left(\begin{smallmatrix} \frac{3}{2} \\ \gamma \end{smallmatrix}\right) \Theta\left(\begin{smallmatrix} \frac{1}{2} \\ \gamma \end{smallmatrix}\right).$$

[Scan p. 511]

Thus taking the tenth line of the upper half, and the fifth line of the lower half, we have

		$\frac{1}{2}\frac{1}{2}$	$\frac{1}{2}\frac{1}{2}$	$\frac{1}{2}\frac{1}{2}$	$\frac{1}{2}\frac{1}{2}$	$\frac{1}{2}\frac{1}{2}$	$\frac{1}{2}\frac{1}{2}$
10	01	$\frac{1}{2}\frac{1}{2}$	$\frac{1}{2}\frac{1}{2}$				
00	10	1 0-1	1 0-1	1 0-1	1 0-1	1 0-1	1 0-1

giving the value of

$$\Theta\left(\begin{smallmatrix} \frac{1}{2} \\ 0 \end{smallmatrix}\right) (u + u') \cdot \Theta\left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right) (u - u');$$

viz. this is

$$\begin{aligned} &= \Theta\left(\begin{smallmatrix} \frac{1}{4} \\ 10 \end{smallmatrix}\right) (2u) \Theta\left(\begin{smallmatrix} \frac{1}{4} \\ -10 \end{smallmatrix}\right) (2u') + \Theta\left(\begin{smallmatrix} \frac{3}{4} \\ 10 \end{smallmatrix}\right) (2u) \Theta\left(\begin{smallmatrix} \frac{1}{4} \\ 10 \end{smallmatrix}\right) (2u') \\ &+ \Theta\left(\begin{smallmatrix} \frac{1}{4} \\ 10 \end{smallmatrix}\right) (\cdot) \Theta\left(\begin{smallmatrix} \frac{3}{4} \\ -10 \end{smallmatrix}\right) (\cdot) - i \Theta\left(\begin{smallmatrix} \frac{1}{4} \\ 10 \end{smallmatrix}\right) (\cdot) \Theta\left(\begin{smallmatrix} \frac{3}{4} \\ 10 \end{smallmatrix}\right) (\cdot) \\ &+ \Theta\left(\begin{smallmatrix} \frac{3}{4} \\ 10 \end{smallmatrix}\right) (\cdot) \Theta\left(\begin{smallmatrix} \frac{1}{4} \\ -10 \end{smallmatrix}\right) (\cdot) + i \Theta\left(\begin{smallmatrix} \frac{3}{4} \\ 10 \end{smallmatrix}\right) (\cdot) \Theta\left(\begin{smallmatrix} \frac{1}{4} \\ 10 \end{smallmatrix}\right) (\cdot) \\ &+ \Theta\left(\begin{smallmatrix} \frac{3}{4} \\ 10 \end{smallmatrix}\right) (\cdot) \Theta\left(\begin{smallmatrix} \frac{3}{4} \\ -10 \end{smallmatrix}\right) (\cdot) - i \Theta\left(\begin{smallmatrix} \frac{3}{4} \\ 10 \end{smallmatrix}\right) (\cdot) \Theta\left(\begin{smallmatrix} \frac{3}{4} \\ 10 \end{smallmatrix}\right) (\cdot), \end{aligned}$$

where the first column is the value given directly by the diagram: it is then reduced to that given by the second column.

84. But instead of the Θ 's, we introduce single letters (X, Y, Z, W) , (E, F, G, H) , (I, J, K, L) , (M, N, P, Q) , with the suffixes $(0, 1, 2, 3)$, in all 64 symbols, thus

$$\Theta \begin{pmatrix} 00 & 10 & 01 & 11 \\ \cdot & \cdot & \cdot & \cdot \end{pmatrix} (2u) = X \ Y \ Z \ W,$$

	00	10	01	11
00	0			
10		1		
01			2	
11				3

that is,

$$\begin{aligned} \Theta_{00}^{00}(2u) &= X, & \Theta_{00}^{10}(2u) &= Y, \text{ \&c.}, \\ \Theta_{01}^{00}(2u) &= Z, \dots, & \Theta_{11}^{00}(2u) &= W. \end{aligned}$$

Similarly,

$$\Theta \begin{pmatrix} \frac{1}{2}0 & \frac{3}{2}0 & \frac{1}{2}1 & \frac{3}{2}1 \\ \cdot & \cdot & \cdot & \cdot \end{pmatrix} (2u) = E \ F \ G \ H,$$

with the suffixes assigned by the same array as above; that is,

$$\Theta \begin{pmatrix} \frac{1}{2}0 \\ \frac{3}{2}0 \\ 00 \end{pmatrix} (2u) = E, \text{ \&c.}$$

[Scan p. 512]

$$\Theta \begin{pmatrix} 0\frac{1}{2} & 1\frac{1}{2} & 0\frac{3}{2} & 1\frac{3}{2} \\ \cdot & \cdot & \cdot & \cdot \end{pmatrix} (2u) = I \ J \ K \ L,$$

with the suffixes $0, 1, 2, 3$ assigned by the same diagonal array as before.

The functions of $(2u')$ are denoted in like manner by accented letters:

$$\Theta_{00}^{00}(2u') = X', \text{ \&c.}$$

$$\Theta \begin{pmatrix} \frac{1}{2}\frac{1}{2} & \frac{3}{2}\frac{1}{2} & \frac{1}{2}\frac{3}{2} & \frac{3}{2}\frac{3}{2} \\ \cdot & \cdot & \cdot & \cdot \end{pmatrix} (2u) = M \ N \ P \ Q,$$

again with the suffixes $0, 1, 2, 3$ assigned by the diagonal array.

85. To simplify the expression of the results, instead of in each case writing down the suffixes, I have indicated them by means of the column headed "Suff."

Thus

$$\left| \begin{array}{c} 8-0 \\ \cdot \end{array} \right| \quad \Theta_{01}^{00}(u+u') \cdot \Theta_{00}^{00}(u-u') = XX' + YY' + ZZ' + WW' \quad \left| \begin{array}{c} 2 \\ \cdot \end{array} \right|$$

means that the equation is to be read

$$= X_2X'_2 + Y_2Y'_2 + Z_2Z'_2 + W_2W'_2.$$

It is hardly necessary to mention that the 8-0 of the left-hand column shows the current numbers of the theta-functions; viz. the left-hand side of the equation is

$$\Theta_8(u+u') \cdot \Theta_0(u-u').$$

And by a preceding remark the single arguments $u+u'$ and $u-u'$ are written in place of $(u+u', v+v')$ and $(u-u', v-v')$ respectively.

[Scan p. 513]

The 256 equations now are

86. First set, 64 equations.

Line	upper	lower	Equation	Suff.
0–0	$\Theta_{00}^{00}u + u'$	$\Theta_{00}^{00}u - u'$	$= XX' + YY' + ZZ' + WW'$	0
4–0	00 / 10	00 / 00	$= XX' + YY' + ZZ' + WW'$	1
8–0	00 / 01	00 / 00	$= XX' + YY' + ZZ' + WW'$	2
12–0	00 / 11	00 / 00	$= XX' + YY' + ZZ' + WW'$	3
0–4	$\Theta_{00}^{00}u + u'$	$\Theta_{10}^{00}u - u'$	$= XX' - YY' + ZZ' - WW'$	1
4–4	00 / 10	00 / 10	$= XX' - YY' + ZZ' - WW'$	0
8–4	00 / 01	00 / 10	$= XX' - YY' + ZZ' - WW'$	3
12–4	00 / 11	00 / 10	$= XX' - YY' + ZZ' - WW'$	2
0–8	$\Theta_{00}^{00}u + u'$	$\Theta_{01}^{00}u - u'$	$= XX' + YY' - ZZ' - WW'$	2
4–8	00 / 10	00 / 01	$= XX' + YY' - ZZ' - WW'$	3
8–8	00 / 01	00 / 01	$= XX' + YY' - ZZ' - WW'$	0
12–8	00 / 11	00 / 01	$= XX' + YY' - ZZ' - WW'$	1
0–12	$\Theta_{00}^{00}u + u'$	$\Theta_{11}^{00}u - u'$	$= XX' - YY' - ZZ' + WW'$	3
4–12	00 / 10	00 / 11	$= XX' - YY' - ZZ' + WW'$	2
8–12	00 / 01	00 / 11	$= XX' - YY' - ZZ' + WW'$	1
12–12	00 / 11	00 / 11	$= XX' - YY' - ZZ' + WW'$	0

[Scan p. 514] First set, 64 equations (continued).

1–1	$\Theta_{00}^{10}u + u'$	$\Theta_{00}^{10}u - u'$	$= YX' + XY' + WZ' + ZW'$	0
5–1	10 / 10	10 / 00	$= YX' + XY' + WZ' + ZW'$	1
9–1	10 / 01	10 / 00	$= YX' + XY' + WZ' + ZW'$	2
13–1	10 / 11	10 / 00	$= YX' + XY' + WZ' + ZW'$	3
1–5	$\Theta_{00}^{10}u + u'$	$\Theta_{10}^{10}u - u'$	$= YX' - XY' + WZ' - ZW'$	1
5–5	10 / 10	10 / 10	$= -YX' + XY' - WZ' + ZW'$	0
9–5	10 / 01	10 / 10	$= YX' - XY' + WZ' - ZW'$	3
13–5	10 / 11	10 / 10	$= -YX' + XY' - WZ' + ZW'$	2
1–9	$\Theta_{00}^{10}u + u'$	$\Theta_{01}^{10}u - u'$	$= YX' + XY' - WZ' - ZW'$	2
5–9	10 / 10	10 / 01	$= YX' + XY' - WZ' - ZW'$	3
9–9	10 / 01	10 / 01	$= YX' + XY' - WZ' - ZW'$	0
13–9	10 / 11	10 / 01	$= YX' + XY' - WZ' - ZW'$	1
1–13	$\Theta_{00}^{10}u + u'$	$\Theta_{11}^{10}u - u'$	$= YX' - XY' - WZ' + ZW'$	3
5–13	10 / 10	10 / 11	$= -YX' + XY' + WZ' - ZW'$	2
9–13	10 / 01	10 / 11	$= YX' - XY' - WZ' + ZW'$	1
13–13	10 / 11	10 / 11	$= -YX' + XY' + WZ' - ZW'$	0

[Scan p. 515] *First set, 64 equations (continued).*

2–2	$\Theta_{00}^{01}u + u'$	$\Theta_{00}^{01}u - u'$	$= ZX' + WY' + XZ' + YW'$	0
6–2	01 / 10	01 / 00	$= ZX' + WY' + XZ' + YW'$	1
10–2	01 / 01	01 / 00	$= ZX' + WY' + XZ' + YW'$	2
14–2	01 / 11	01 / 00	$= ZX' + WY' + XZ' + YW'$	3
2–6	$\Theta_{00}^{01}u + u'$	$\Theta_{10}^{01}u - u'$	$= ZX' - WY' + XZ' - YW'$	1
6–6	01 / 10	01 / 10	$= ZX' - WY' + XZ' - YW'$	0
10–6	01 / 01	01 / 10	$= ZX' - WY' + XZ' - YW'$	3
14–6	01 / 11	01 / 10	$= ZX' - WY' + XZ' - YW'$	2
2–10	$\Theta_{00}^{01}u + u'$	$\Theta_{01}^{01}u - u'$	$= ZX' + WY' - XZ' - YW'$	2
6–10	01 / 10	01 / 01	$= ZX' + WY' - XZ' - YW'$	3
10–10	01 / 01	01 / 01	$= -ZX' - WY' + XZ' + YW'$	0
14–10	01 / 11	01 / 01	$= -ZX' - WY' + XZ' + YW'$	1
2–14	$\Theta_{00}^{01}u + u'$	$\Theta_{11}^{01}u - u'$	$= ZX' - WY' - XZ' + YW'$	3
6–14	01 / 10	01 / 11	$= ZX' - WY' - XZ' + YW'$	2
10–14	01 / 01	01 / 11	$= -ZX' + WY' + XZ' - YW'$	1
14–14	01 / 11	01 / 11	$= -ZX' + WY' + XZ' - YW'$	0

[Scan p. 516] *First set, 64 equations (concluded).*

3–3	$\Theta_{00}^{11}u + u'$	$\Theta_{00}^{11}u - u'$	$= WX' + ZY' + YZ' + XW'$	0
7–3	11 / 10	11 / 00	$= WX' + ZY' + YZ' + XW'$	1
11–3	11 / 01	11 / 00	$= WX' + ZY' + YZ' + XW'$	2
15–3	11 / 11	11 / 00	$= WX' + ZY' + YZ' + XW'$	3
3–7	$\Theta_{00}^{11}u + u'$	$\Theta_{10}^{11}u - u'$	$= WX' - ZY' + YZ' - XW'$	1
7–7	11 / 10	11 / 10	$= -WX' + ZY' - YZ' + XW'$	0
11–7	11 / 01	11 / 10	$= WX' - ZY' + YZ' - XW'$	3
15–7	11 / 11	11 / 10	$= -WX' + ZY' - YZ' + XW'$	2
3–11	$\Theta_{00}^{11}u + u'$	$\Theta_{01}^{11}u - u'$	$= WX' + ZY' - YZ' - XW'$	2
7–11	11 / 10	11 / 01	$= WX' + ZY' - YZ' - XW'$	3
11–11	11 / 01	11 / 01	$= -WX' - ZY' + YZ' + XW'$	0
15–11	11 / 11	11 / 01	$= -WX' - ZY' + YZ' + XW'$	1
3–15	$\Theta_{00}^{11}u + u'$	$\Theta_{11}^{11}u - u'$	$= WX' - ZY' - YZ' + XW'$	3
7–15	11 / 10	11 / 11	$= -WX' + ZY' + YZ' - XW'$	2
11–15	11 / 01	11 / 11	$= -WX' + ZY' + YZ' - XW'$	1
15–15	11 / 11	11 / 11	$= WX' - ZY' - YZ' + XW'$	0

[Scan p. 517]

87. Second set, 64 equations.

1–0	$\Theta_{00}^{10}u + u'$	$\Theta_{00}^{00}u - u'$	$= EE' + GG' + FF' + HH'$	0
5–0	10 / 10	00 / 00	$= EE' + GG' + FF' + HH'$	1
9–0	10 / 01	00 / 00	$= EE' + GG' + FF' + HH'$	2
13–0	10 / 11	00 / 00	$= EE' + GG' + FF' + HH'$	3
1–4	$\Theta_{00}^{10}u + u'$	$\Theta_{10}^{00}u - u'$	$= -iEE' + iGG' - iFF' + iHH'$	1
5–4	10 / 10	00 / 10	$= iEE' - iGG' + iFF' - iHH'$	0
9–4	10 / 01	00 / 10	$= -iEE' + iGG' - iFF' + iHH'$	3
13–4	10 / 11	00 / 10	$= iEE' - iGG' + iFF' - iHH'$	2
1–8	$\Theta_{00}^{10}u + u'$	$\Theta_{01}^{00}u - u'$	$= EE' + GG' - FF' - HH'$	2
5–8	10 / 10	00 / 01	$= EE' + GG' - FF' - HH'$	3
9–8	10 / 01	00 / 01	$= EE' + GG' - FF' - HH'$	0
13–8	10 / 11	00 / 01	$= EE' + GG' - FF' - HH'$	1
1–12	$\Theta_{00}^{10}u + u'$	$\Theta_{11}^{00}u - u'$	$= -iEE' + iGG' + iFF' - iHH'$	3
5–12	10 / 10	00 / 11	$= iEE' - iGG' - iFF' + iHH'$	2
9–12	10 / 01	00 / 11	$= -iEE' + iGG' + iFF' - iHH'$	1
13–12	10 / 11	00 / 11	$= iEE' - iGG' - iFF' + iHH'$	0

[Scan p. 518] Second set, 64 equations (continued).

0–1	$\Theta_{00}^{00}u + u'$	$\Theta_{00}^{10}u - u'$	$= EG' + GE' + FH' + HF'$	0
4–1	00 / 10	10 / 00	$= EG' + GE' + FH' + HF'$	1
8–1	00 / 01	10 / 00	$= EG' + GE' + FH' + HF'$	2
12–1	00 / 11	10 / 00	$= EG' + GE' + FH' + HF'$	3
0–5	$\Theta_{00}^{00}u + u'$	$\Theta_{10}^{10}u - u'$	$= iEG' - iGE' + iFH' - iHF'$	1
4–5	00 / 10	10 / 10	$= iEG' - iGE' + iFH' - iHF'$	0
8–5	00 / 01	10 / 10	$= iEG' - iGE' + iFH' - iHF'$	3
12–5	00 / 11	10 / 10	$= iEG' - iGE' + iFH' - iHF'$	2
0–9	$\Theta_{00}^{00}u + u'$	$\Theta_{01}^{10}u - u'$	$= EG' + GE' - FH' - HF'$	2
4–9	00 / 10	10 / 01	$= EG' + GE' - FH' - HF'$	3
8–9	00 / 01	10 / 01	$= EG' + GE' - FH' - HF'$	0
12–9	00 / 11	10 / 01	$= EG' + GE' - FH' - HF'$	1
0–13	$\Theta_{00}^{00}u + u'$	$\Theta_{11}^{10}u - u'$	$= iEG' - iGE' - iFH' + iHF'$	3
4–13	00 / 10	10 / 11	$= iEG' - iGE' - iFH' + iHF'$	2
8–13	00 / 01	10 / 11	$= iEG' - iGE' - iFH' + iHF'$	1
12–13	00 / 11	10 / 11	$= iEG' - iGE' - iFH' + iHF'$	0

[Scan p. 519] Second set, 64 equations (continued).

3–2	$\Theta_{00}^{11}u + u'$	$\Theta_{00}^{01}u - u'$	$= FE' + HG' + EF' + GH'$	0
7–2	11 / 10	01 / 00	$= FE' + HG' + EF' + GH'$	1
11–2	11 / 01	01 / 00	$= FE' + HG' + EF' + GH'$	2
15–2	11 / 11	01 / 00	$= FE' + HG' + EF' + GH'$	3
3–6	$\Theta_{00}^{11}u + u'$	$\Theta_{10}^{01}u - u'$	$= -iFE' + iHG' - iEF' + iGH'$	1
7–6	11 / 10	01 / 10	$= iFE' - iHG' + iEF' - iGH'$	0
11–6	11 / 01	01 / 10	$= -iFE' + iHG' - iEF' + iGH'$	3
15–6	11 / 11	01 / 10	$= iFE' - iHG' + iEF' - iGH'$	2
3–10	$\Theta_{00}^{11}u + u'$	$\Theta_{01}^{01}u - u'$	$= FE' + HG' - EF' - GH'$	2
7–10	11 / 10	01 / 01	$= FE' + HG' - EF' - GH'$	3
11–10	11 / 01	01 / 01	$= -FE' - HG' + EF' + GH'$	0
15–10	11 / 11	01 / 01	$= -FE' - HG' + EF' + GH'$	1
3–14	$\Theta_{00}^{11}u + u'$	$\Theta_{11}^{01}u - u'$	$= -iFE' + iHG' + iEF' - iGH'$	3
7–14	11 / 10	01 / 11	$= iFE' - iHG' - iEF' + iGH'$	2
11–14	11 / 01	01 / 11	$= iFE' - iHG' - iEF' + iGH'$	1
15–14	11 / 11	01 / 11	$= -iFE' + iHG' + iEF' - iGH'$	0

[Scan p. 520] *Second set, 64 equations (concluded).*

2–3	$\Theta_{00}^{01}u + u'$	$\Theta_{00}^{11}u - u'$	$= FG' + HE' + EH' + GF'$	0
6–3	01 / 10	11 / 00	$= FG' + HE' + EH' + GF'$	1
10–3	01 / 01	11 / 00	$= FG' + HE' + EH' + GF'$	2
14–3	01 / 11	11 / 00	$= FG' + HE' + EH' + GF'$	3
2–7	$\Theta_{00}^{01}u + u'$	$\Theta_{10}^{11}u - u'$	$= iFG' - iHE' + iEH' - iGF'$	1
6–7	01 / 10	11 / 10	$= iFG' - iHE' + iEH' - iGF'$	0
10–7	01 / 01	11 / 10	$= iFG' - iHE' + iEH' - iGF'$	3
14–7	01 / 11	11 / 10	$= iFG' - iHE' + iEH' - iGF'$	2
2–11	$\Theta_{00}^{01}u + u'$	$\Theta_{01}^{11}u - u'$	$= FG' + HE' - EH' - GF'$	2
6–11	01 / 10	11 / 01	$= FG' + HE' - EH' - GF'$	3
10–11	01 / 01	11 / 01	$= -FG' - HE' + EH' + GF'$	0
14–11	01 / 11	11 / 01	$= -FG' - HE' + EH' + GF'$	1
2–15	$\Theta_{00}^{01}u + u'$	$\Theta_{11}^{11}u - u'$	$= iFG' - iHE' - iEH' + iGF'$	3
6–15	01 / 10	11 / 11	$= iFG' - iHE' - iEH' + iGF'$	2
10–15	01 / 01	11 / 11	$= -iFG' + iHE' + iEH' - iGF'$	1
14–15	01 / 11	11 / 11	$= -iFG' + iHE' + iEH' - iGF'$	0

[Scan p. 521]

2–0	$\Theta_{00}^{01}u + u'$	$\Theta_{00}^{00}u - u'$	$= II' + JJ' + KK' + LL'$	0
6–0	01 / 10	00 / 00	$= II' + JJ' + KK' + LL'$	1
10–0	01 / 01	00 / 00	$= II' + JJ' + KK' + LL'$	2
14–0	01 / 11	00 / 00	$= II' + JJ' + KK' + LL'$	3
2–4	$\Theta_{00}^{01}u + u'$	$\Theta_{10}^{00}u - u'$	$= II' - JJ' + KK' - LL'$	1
6–4	01 / 10	00 / 10	$= II' - JJ' + KK' - LL'$	0
10–4	01 / 01	00 / 10	$= II' - JJ' + KK' - LL'$	3
14–4	01 / 11	00 / 10	$= II' - JJ' + KK' - LL'$	2
2–8	$\Theta_{00}^{01}u + u'$	$\Theta_{01}^{00}u - u'$	$= -iII' - iJJ' + iKK' + iLL'$	2
6–8	01 / 10	00 / 01	$= -iII' - iJJ' + iKK' + iLL'$	3
10–8	01 / 01	00 / 01	$= iII' + iJJ' - iKK' - iLL'$	0
14–8	01 / 11	00 / 01	$= iII' + iJJ' - iKK' - iLL'$	1
2–12	$\Theta_{00}^{01}u + u'$	$\Theta_{11}^{00}u - u'$	$= -iII' + iJJ' + iKK' - iLL'$	3
6–12	01 / 10	00 / 11	$= -iII' + iJJ' + iKK' - iLL'$	2
10–12	01 / 01	00 / 11	$= iII' - iJJ' - iKK' + iLL'$	1
14–12	01 / 11	00 / 11	$= iII' - iJJ' - iKK' + iLL'$	0

[Scan p. 522] Third set, 64 equations (continued).

3–1	$\Theta_{00}^{11}u + u'$	$\Theta_{00}^{10}u - u'$	$= JJ' + IJ' + LK' + KL'$	0
7–1	11 / 10	10 / 00	$= JJ' + IJ' + LK' + KL'$	1
11–1	11 / 01	10 / 00	$= JJ' + IJ' + LK' + KL'$	2
15–1	11 / 11	10 / 00	$= JJ' + IJ' + LK' + KL'$	3
3–5	$\Theta_{00}^{11}u + u'$	$\Theta_{10}^{10}u - u'$	$= JJ' - IJ' + LK' - KL'$	1
7–5	11 / 10	10 / 10	$= -JJ' + IJ' - LK' + KL'$	0
11–5	11 / 01	10 / 10	$= JJ' - IJ' + LK' - KL'$	3
15–5	11 / 11	10 / 10	$= -JJ' + IJ' - LK' + KL'$	2
3–9	$\Theta_{00}^{11}u + u'$	$\Theta_{01}^{10}u - u'$	$= -iJJ' - iIJ' + iLK' + iKL'$	2
7–9	11 / 10	10 / 01	$= -iJJ' - iIJ' + iLK' + iKL'$	3
11–9	11 / 01	10 / 01	$= iJJ' + iIJ' - iLK' - iKL'$	0
15–9	11 / 11	10 / 01	$= iJJ' + iIJ' - iLK' - iKL'$	1
3–13	$\Theta_{00}^{11}u + u'$	$\Theta_{11}^{10}u - u'$	$= -iJJ' + iIJ' + iLK' - iKL'$	3
7–13	11 / 10	10 / 11	$= iJJ' - iIJ' - iLK' + iKL'$	2
11–13	11 / 01	10 / 11	$= iJJ' - iIJ' - iLK' + iKL'$	1
15–13	11 / 11	10 / 11	$= -iJJ' + iIJ' + iLK' - iKL'$	0

[Scan p. 523] Third set, 64 equations (continued).

0–2	$\Theta_{00}^{00}u + u'$	$\Theta_{00}^{01}u - u'$	$= IK' + JL' + KI' + LJ'$	0
4–2	00 / 10	01 / 00	$= IK' + JL' + KI' + LJ'$	1
8–2	00 / 01	01 / 00	$= IK' + JL' + KI' + LJ'$	2
12–2	00 / 11	01 / 00	$= IK' + JL' + KI' + LJ'$	3
0–6	$\Theta_{00}^{00}u + u'$	$\Theta_{10}^{01}u - u'$	$= IK' - JL' + KI' - LJ'$	1
4–6	00 / 10	01 / 10	$= IK' - JL' + KI' - LJ'$	0
8–6	00 / 01	01 / 10	$= IK' - JL' + KI' - LJ'$	3
12–6	00 / 11	01 / 10	$= IK' - JL' + KI' - LJ'$	2
0–10	$\Theta_{00}^{00}u + u'$	$\Theta_{01}^{01}u - u'$	$= iIK' + iJL' - iKI' - iLJ'$	2
4–10	00 / 10	01 / 01	$= iIK' + iJL' - iKI' - iLJ'$	3
8–10	00 / 01	01 / 01	$= iIK' + iJL' - iKI' - iLJ'$	0
12–10	00 / 11	01 / 01	$= iIK' + iJL' - iKI' - iLJ'$	1
0–14	$\Theta_{00}^{00}u + u'$	$\Theta_{11}^{01}u - u'$	$= iIK' - iJL' - iKI' + iLJ'$	3
4–14	00 / 10	01 / 11	$= iIK' - iJL' - iKI' + iLJ'$	2
8–14	00 / 01	01 / 11	$= iIK' - iJL' - iKI' + iLJ'$	1
12–14	00 / 11	01 / 11	$= iIK' - iJL' - iKI' + iLJ'$	0

[Scan p. 524] *Third set, 64 equations (concluded).*

1–3	$\Theta_{00}^{10}u + u'$	$\Theta_{00}^{11}u - u'$	$= JK' + IL' + LI' + KJ'$	0
5–3	10 / 10	11 / 00	$= JK' + IL' + LI' + KJ'$	1
9–3	10 / 01	11 / 00	$= JK' + IL' + LI' + KJ'$	2
13–3	10 / 11	11 / 00	$= JK' + IL' + LI' + KJ'$	3
1–7	$\Theta_{00}^{10}u + u'$	$\Theta_{10}^{11}u - u'$	$= JK' - IL' + LI' - KJ'$	1
5–7	10 / 10	11 / 10	$= -JK' + IL' - LI' + KJ'$	0
9–7	10 / 01	11 / 10	$= JK' - IL' + LI' - KJ'$	3
13–7	10 / 11	11 / 10	$= -JK' + IL' - LI' + KJ'$	2
1–11	$\Theta_{00}^{10}u + u'$	$\Theta_{01}^{11}u - u'$	$= iJK' + iIL' - iLI' - iKJ'$	2
5–11	10 / 10	11 / 01	$= iJK' + iIL' - iLI' - iKJ'$	3
9–11	10 / 01	11 / 01	$= iJK' + iIL' - iLI' - iKJ'$	0
13–11	10 / 11	11 / 01	$= iJK' + iIL' - iLI' - iKJ'$	1
1–15	$\Theta_{00}^{10}u + u'$	$\Theta_{11}^{11}u - u'$	$= iJK' - iIL' - iLI' + iKJ'$	3
5–15	10 / 10	11 / 11	$= -iJK' + iIL' + iLI' - iKJ'$	2
9–15	10 / 01	11 / 11	$= iJK' - iIL' - iLI' + iKJ'$	1
13–15	10 / 11	11 / 11	$= -iJK' + iIL' + iLI' - iKJ'$	0

[Scan p. 525]

89. Fourth set, 64 equations.

3–0	$\Theta_{00}^{11}u + u'$	$\Theta_{00}^{00}u - u'$	$= MM' + NN' + PP' + QQ'$	0
7–0	11 / 10	00 / 00	$= MM' + NN' + PP' + QQ'$	1
11–0	11 / 01	00 / 00	$= MM' + NN' + PP' + QQ'$	2
15–0	11 / 11	00 / 00	$= MM' + NN' + PP' + QQ'$	3
3–4	$\Theta_{00}^{11}u + u'$	$\Theta_{10}^{00}u - u'$	$= -iMM' + iNN' - iPP' + iQQ'$	1
7–4	11 / 10	00 / 10	$= +iMM' - iNN' + iPP' - iQQ'$	0
11–4	11 / 01	00 / 10	$= -iMM' + iNN' - iPP' + iQQ'$	3
15–4	11 / 11	00 / 10	$= +iMM' - iNN' + iPP' - iQQ'$	2
3–8	$\Theta_{00}^{11}u + u'$	$\Theta_{01}^{00}u - u'$	$= -iMM' - iNN' + iPP' + iQQ'$	2
7–8	11 / 10	00 / 01	$= -iMM' - iNN' + iPP' + iQQ'$	3
11–8	11 / 01	00 / 01	$= iMM' + iNN' - iPP' - iQQ'$	0
15–8	11 / 11	00 / 01	$= iMM' + iNN' - iPP' - iQQ'$	1
3–12	$\Theta_{00}^{11}u + u'$	$\Theta_{11}^{00}u - u'$	$= -MM' + NN' + PP' - QQ'$	3
7–12	11 / 10	00 / 11	$= +MM' - NN' - PP' + QQ'$	2
11–12	11 / 01	00 / 11	$= +MM' - NN' - PP' + QQ'$	1
15–12	11 / 11	00 / 11	$= -MM' + NN' + PP' - QQ'$	0

[Scan p. 526] Fourth set, 64 equations (continued).

2–1	$\Theta_{00}^{01}u + u'$	$\Theta_{00}^{10}u - u'$	$= MN' + NM' + PQ' + QP'$	0
6–1	01 / 10	10 / 00	$= MN' + NM' + PQ' + QP'$	1
10–1	01 / 01	10 / 00	$= MN' + NM' + PQ' + QP'$	2
14–1	01 / 11	10 / 00	$= MN' + NM' + PQ' + QP'$	3
2–5	$\Theta_{00}^{01}u + u'$	$\Theta_{10}^{10}u - u'$	$= iMN' - iNM' + iPQ' - iQP'$	1
6–5	01 / 10	10 / 10	$= iMN' - iNM' + iPQ' - iQP'$	0
10–5	01 / 01	10 / 10	$= iMN' - iNM' + iPQ' - iQP'$	3
14–5	01 / 11	10 / 10	$= iMN' - iNM' + iPQ' - iQP'$	2
2–9	$\Theta_{00}^{01}u + u'$	$\Theta_{01}^{10}u - u'$	$= -iMN' - iNM' + iPQ' + iQP'$	2
6–9	01 / 10	10 / 01	$= -iMN' - iNM' + iPQ' + iQP'$	3
10–9	01 / 01	10 / 01	$= iMN' + iNM' - iPQ' - iQP'$	0
14–9	01 / 11	10 / 01	$= iMN' + iNM' - iPQ' - iQP'$	1
2–13	$\Theta_{00}^{01}u + u'$	$\Theta_{11}^{10}u - u'$	$= MN' - NM' - PQ' + QP'$	3
6–13	01 / 10	10 / 11	$= MN' - NM' - PQ' + QP'$	2
10–13	01 / 01	10 / 11	$= -MN' + NM' + PQ' - QP'$	1
14–13	01 / 11	10 / 11	$= -MN' + NM' + PQ' - QP'$	0

[Scan p. 527] Fourth set, 64 equations (continued).

1–2	$\Theta_{00}^{10}u + u'$	$\Theta_{00}^{01}u - u'$	$= MP' + NQ' + PM' + QN'$	0
5–2	10 / 10	01 / 00	$= MP' + NQ' + PM' + QN'$	1
9–2	10 / 01	01 / 00	$= MP' + NQ' + PM' + QN'$	2
13–2	10 / 11	01 / 00	$= MP' + NQ' + PM' + QN'$	3
1–6	$\Theta_{00}^{10}u + u'$	$\Theta_{10}^{01}u - u'$	$= -iMP' + iNQ' - iPM' + iQN'$	1
5–6	10 / 10	01 / 10	$= iMP' - iNQ' + iPM' - iQN'$	0
9–6	10 / 01	01 / 10	$= -iMP' + iNQ' - iPM' + iQN'$	3
13–6	10 / 11	01 / 10	$= iMP' - iNQ' + iPM' - iQN'$	2
1–10	$\Theta_{00}^{10}u + u'$	$\Theta_{01}^{01}u - u'$	$= iMP' + iNQ' - iPM' - iQN'$	2
5–10	10 / 10	01 / 01	$= iMP' + iNQ' - iPM' - iQN'$	3
9–10	10 / 01	01 / 01	$= iMP' + iNQ' - iPM' - iQN'$	0
13–10	10 / 11	01 / 01	$= iMP' + iNQ' - iPM' - iQN'$	1
1–14	$\Theta_{00}^{10}u + u'$	$\Theta_{11}^{01}u - u'$	$= MP' - NQ' - PM' + QN'$	3
5–14	10 / 10	01 / 11	$= -MP' + NQ' + PM' - QN'$	2
9–14	10 / 01	01 / 11	$= MP' - NQ' - PM' + QN'$	1
13–14	10 / 11	01 / 11	$= -MP' + NQ' + PM' - QN'$	0

[Scan p. 528] *Fourth set, 64 equations (concluded).*

0–3	$\Theta_{00}^{00}u + u'$	$\Theta_{00}^{11}u - u'$	$= MQ' + NP' + PN' + QM'$	0
4–3	00 / 10	11 / 00	$= MQ' + NP' + PN' + QM'$	1
8–3	00 / 01	11 / 00	$= MQ' + NP' + PN' + QM'$	2
12–3	00 / 11	11 / 00	$= MQ' + NP' + PN' + QM'$	3
0–7	$\Theta_{00}^{00}u + u'$	$\Theta_{10}^{11}u - u'$	$= iMQ' - iNP' + iPN' - iQM'$	1
4–7	00 / 10	11 / 10	$= iMQ' - iNP' + iPN' - iQM'$	0
8–7	00 / 01	11 / 10	$= iMQ' - iNP' + iPN' - iQM'$	3
12–7	00 / 11	11 / 10	$= iMQ' - iNP' + iPN' - iQM'$	2
0–11	$\Theta_{00}^{00}u + u'$	$\Theta_{01}^{11}u - u'$	$= iMQ' + iNP' - iPN' - iQM'$	2
4–11	00 / 10	11 / 01	$= iMQ' + iNP' - iPN' - iQM'$	3
8–11	00 / 01	11 / 01	$= iMQ' + iNP' - iPN' - iQM'$	0
12–11	00 / 11	11 / 01	$= iMQ' + iNP' - iPN' - iQM'$	1
0–15	$\Theta_{00}^{00}u + u'$	$\Theta_{11}^{11}u - u'$	$= -MQ' + NP' + PN' - QM'$	3
4–15	00 / 10	11 / 11	$= -MQ' + NP' + PN' - QM'$	2
8–15	00 / 01	11 / 11	$= -MQ' + NP' + PN' - QM'$	1
12–15	00 / 11	11 / 11	$= -MQ' + NP' + PN' - QM'$	0

[Scan p. 529]

90. I re-arrange these in sets of 16 equations, the equations of the first or square-set of 16 being taken as they stand, but those of the other sets being combined in pairs by addition and subtraction as will be

seen. And I now drop altogether the characteristics, retaining only the current numbers: thus, in the set of equations next written down, the first equation is

$$\Theta_0(u + u') \Theta_0(u - u') = XX' + YY' + ZZ' + WW';$$

in the second set, the first equation is

$$\tfrac{1}{2}[\Theta_0(u + u') \Theta_4(u - u') + \Theta_4(u + u') \Theta_0(u - u')] = X_1X'_1 + Z_1Z'_1,$$

and so in other cases.

First or square-set of 16.

$u + u'$	$u - u'$	Equation	(Suffixes 0)
0	0	$= XX' + YY' + ZZ' + WW'$	
4	4	$= XX' - YY' + ZZ' - WW'$	
8	8	$= XX' + YY' - ZZ' - WW'$	
12	12	$= XX' - YY' - ZZ' + WW'$	
1	1	$= YX' + XY' + WZ' + ZW'$	
5	5	$= -YX' + XY' - WZ' + ZW'$	
9	9	$= YX' + XY' - WZ' - ZW'$	
13	13	$= -YX' + XY' + WZ' - ZW'$	
2	2	$= ZX' + WY' + XZ' + YW'$	
6	6	$= ZX' - WY' + XZ' - YW'$	
10	10	$= ZX' - WY' - XZ' + YW'$	
14	14	$= -ZX' + WY' + XZ' - YW'$	
3	3	$= WX' + ZY' + YZ' + XW'$	
7	7	$= -WX' + ZY' - YZ' + XW'$	
11	11	$= -WX' - ZY' + YZ' + XW'$	
15	15	$= WX' - ZY' - YZ' + XW'$	

91. Second set of 16.

$\tfrac{1}{2}\{u + u' \quad u - u' \quad u + u' \quad u - u'\}$	Equation	(Suffixes 1.)
4 0 0 4	$= XX' + ZZ'$	
12 8 8 12	$= XX' - ZZ'$	
5 1 1 5	$= YX' + WZ'$	
13 9 9 13	$= YX' - WZ'$	
6 2 2 6	$= ZX' + XZ'$	
14 10 10 14	$= ZX' + XZ'$	
7 3 3 7	$= WX' + YZ'$	
15 11 11 15	$= WX' + YZ'$	

$\frac{1}{2}\{u+u' \quad u-u' \quad u+u' \quad u-u'\}$	Equation	(Suffixes 1.)
4 0 0 4	$= YY' + WW'$	
12 8 8 12	$= -YY' + WW'$	
5 1 1 5	$= XY' + ZW'$	
13 9 9 13	$= -XY' + ZW'$	
6 2 2 6	$= WY' + YW'$	
14 10 10 14	$= WY' - YW'$	
7 3 3 7	$= ZY' + XW'$	
15 11 11 15	$= -ZY' + XW'$	

[Scan p. 530]

92. Third set of 16.

$\frac{1}{2}\{u+u' \quad u-u' \quad u+u' \quad u-u'\}$	Equation	(Suffixes 2.)
8 0 0 8	$= XX' + YY'$	
12 4 4 12	$= XX' - YY'$	
9 1 1 9	$= YX' + XY'$	
13 5 5 13	$= -YX' + XY'$	
10 2 2 10	$= ZX' + WY'$	
14 6 6 14	$= ZX' - WY'$	
11 3 3 11	$= WX' + ZY'$	
15 7 7 15	$= -WX' + ZY'$	

$\frac{1}{2}\{u+u' \quad u-u' \quad u+u' \quad u-u'\}$	Equation	(Suffixes 2.)
8 0 0 8	$= ZZ' + WW'$	
12 4 4 12	$= ZZ' - WW'$	
9 1 1 9	$= WZ' + ZW'$	
13 5 5 13	$= -WZ' + ZW'$	
10 2 2 10	$= XZ' + YW'$	
14 6 6 14	$= XZ' - YW'$	
11 3 3 11	$= YZ' + XW'$	
15 7 7 15	$= -YZ' + XW'$	

93. Fourth set of 16.

$\frac{1}{2}\{u+u' \quad u-u' \quad u+u' \quad u-u'\}$	Equation	(Suffixes 3.)
12	$= XX' + WW'$	
8	$= XX' - WW'$	
13	$= YX' + ZW'$	
9	$= YX' - ZW'$	
14	$= ZX' + YW'$	
10	$= ZX' - YW'$	
15	$= WX' + XW'$	
11	$= WX' - XW'$	

$\frac{1}{2}\{u+u' \quad u-u' \quad u+u' \quad u-u'\}$	Equation	(Suffixes 3.)
12	$= YY' + ZZ'$	
8	$= -YY' + ZZ'$	
13	$= XY' + WZ'$	
9	$= -XY' + WZ'$	
14	$= WY' + XZ'$	
10	$= -WY' + XZ'$	
15	$= ZY' + YZ'$	
11	$= -ZY' + YZ'$	